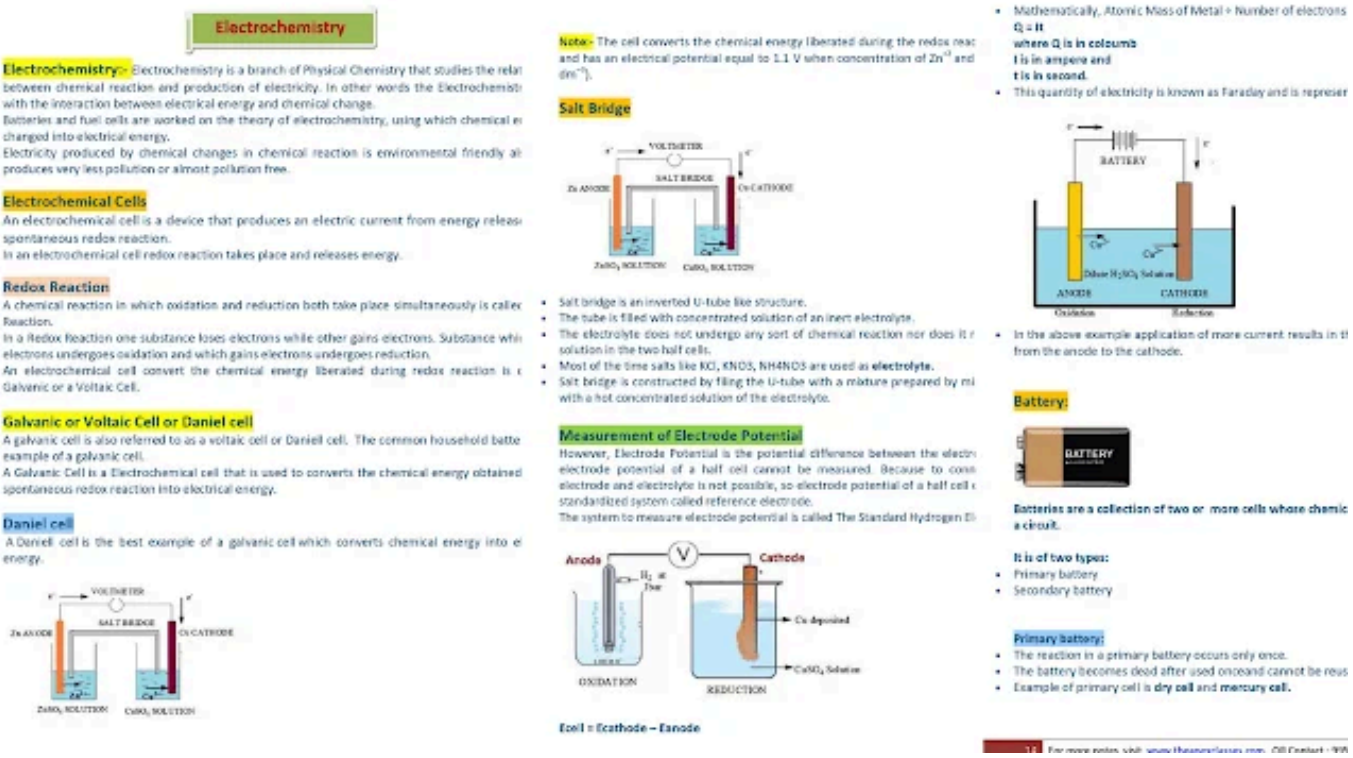

Electrochemistry

Electrochemistry:- Electrochemistry is a branch of Physical Chemistry that studies the relat between chemical reaction and production of electricity. In other words the Electrochemist with the interaction between electrical energy and chemical change.
Batteries and fuel cells are worked on the theory of electrochemistry, using which chemical e changed into electrical energy.
Electricity produced by chemical changes in chemical reaction is environmental friendly al produces very less pollution or almost pollution free.

Electrochemical Cells
An electrochemical cell is a device that produces an electric current from energy releas spontaneous redox reaction.
In an electrochemical cell redox reaction takes place and releases energy.

Redox Reaction
A chemical reaction in which oxidation and reduction both take place simultaneously is calle Reaction.
In a Redox Reaction one substance loses electrons while other gains electrons. Substance whi electrons undergoes oxidation and which gains electrons undergoes reduction.
An electrochemical cell convert the chemical energy liberated during redox reaction is Galvanic or a Voltaic Cell.

Galvanic or Voltaic Cell or Daniel cell
A galvanic cell is also referred to as a voltaic cell or Daniell cell. The common household batte example of a galvanic cell.
A Galvanic Cell is a Electrochemical cell that is used to converts the chemical energy obtained spontaneous redox reaction into electrical energy.

Daniel cell
A Daniell cell is the best example of a galvanic cell which converts chemical energy into e energy.

VOLTMETER
SALT BRIDGE
Zn ANODE
Cu CATHODE
$ZnSO_4$ SOLUTION
$CuSO_4$ SOLUTION

Note:- The cell converts the chemical energy liberated during the redox reac and has an electrical potential equal to 1.1 V when concentration of Zn^{+2} and dm^{-3}).

Salt Bridge

VOLTMETER
SALT BRIDGE
Zn ANODE
Cu CATHODE
$ZnSO_4$ SOLUTION
$CuSO_4$ SOLUTION

- Salt bridge is an inverted U-tube like structure.
- The tube is filled with concentrated solution of an inert electrolyte.
- The electrolyte does not undergo any sort of chemical reaction nor does it r solution in the two half cells.
- Most of the time salts like KCl, KNO3, NH4NO3 are used as **electrolyte.**
- Salt bridge is constructed by filing the U-tube with a mixture prepared by mi with a hot concentrated solution of the electrolyte.

Measurement of Electrode Potential
However, Electrode Potential is the potential difference between the electr electrode potential of a half cell cannot be measured. Because to conn electrode and electrolyte is not possible, so electrode potential of a half cell standardized system called reference electrode.
The system to measure electrode potential is called The Standard Hydrogen El

Anode
V
Cathode
Cu deposited
$CuSO_4$ Solution
OXIDATION
REDUCTION

Ecell = Ecathode – Eanode

- Mathematically, Atomic Mass of Metal × Number of electrons
  Q = It
  where Q is in coloumb
  I is in ampere and
  t is in second.
- This quantity of electricity is known as Faraday and is represer

BATTERY
ANODE
CATHODE
Oxidation
Reduction

- In the above example application of more current results in t from the anode to the cathode.

Battery:

BATTERY

Batteries are a collection of two or more cells whose chemic a circuit.

It is of two types:
- Primary battery
- Secondary battery

Primary battery:
- The reaction in a primary battery occurs only once.
- The battery becomes dead after used onceand cannot be reus
- Example of primary cell is **dry cell** and **mercury cell.**

Class 12 Chemistry Notes Chapter 3 Electrochemistry

Class 12 Chemistry Notes

Class 12 Chemistry Notes Chapter 3 Electrochemistry

Electrochemistry is an important section of chemistry that determines the function of electrodes and reactors. Notesstreet Electrochemistry Notes Class 12 attempts to establish the ideas behind chemical reactions.

An electrochemical cell is a device that produces differences between electrode forms through a chemical reaction. Ideally there are two types of electron conductors which are separated by an ionic conductor. An electron conductor connects it further, making it accessible.

Notes for class 12 chemistry chapter 3 explain this function of electrons where two metal electrodes are present. These metal electrodes are immersed in an electrolytic solution to produce electricity. By thoroughly reading Chapter 3 Chemistry Class 12 Notes students will come to know that ionic conductors are an important part of cells.

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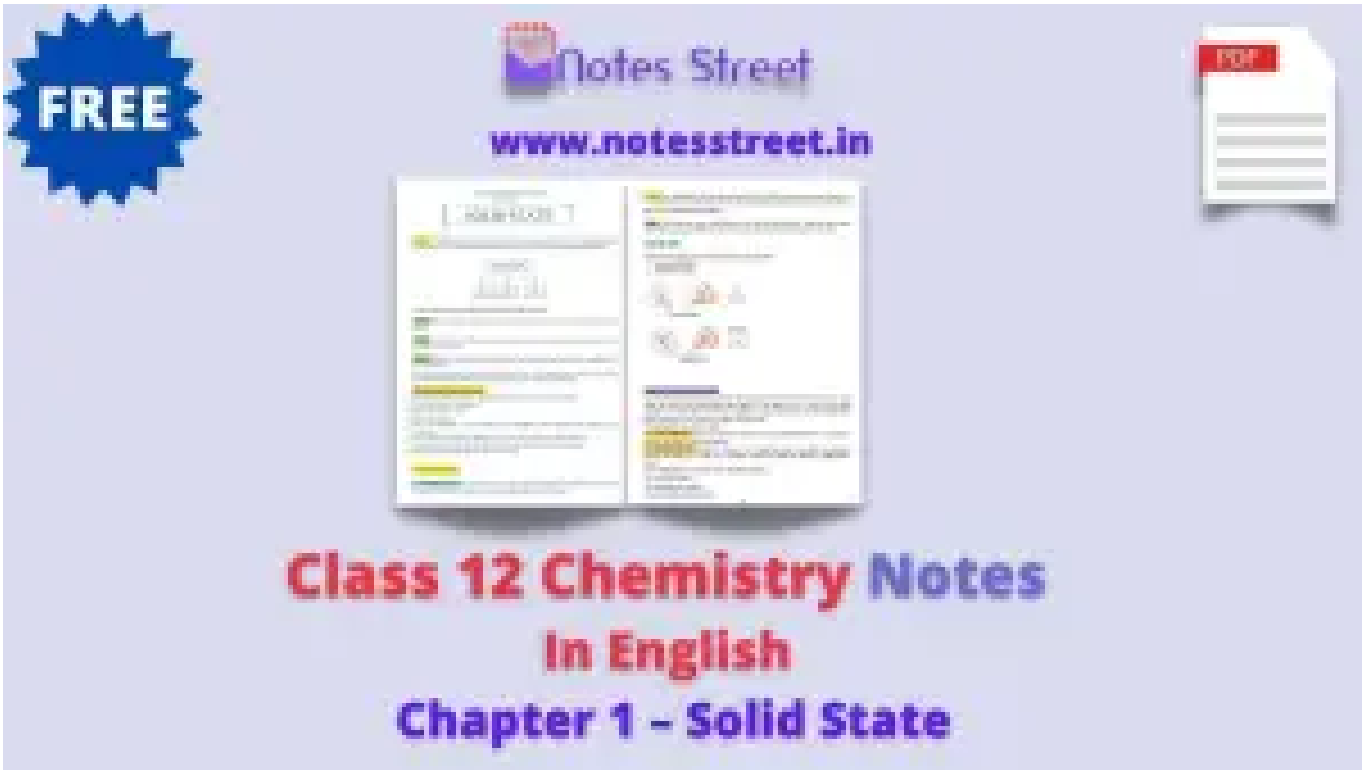
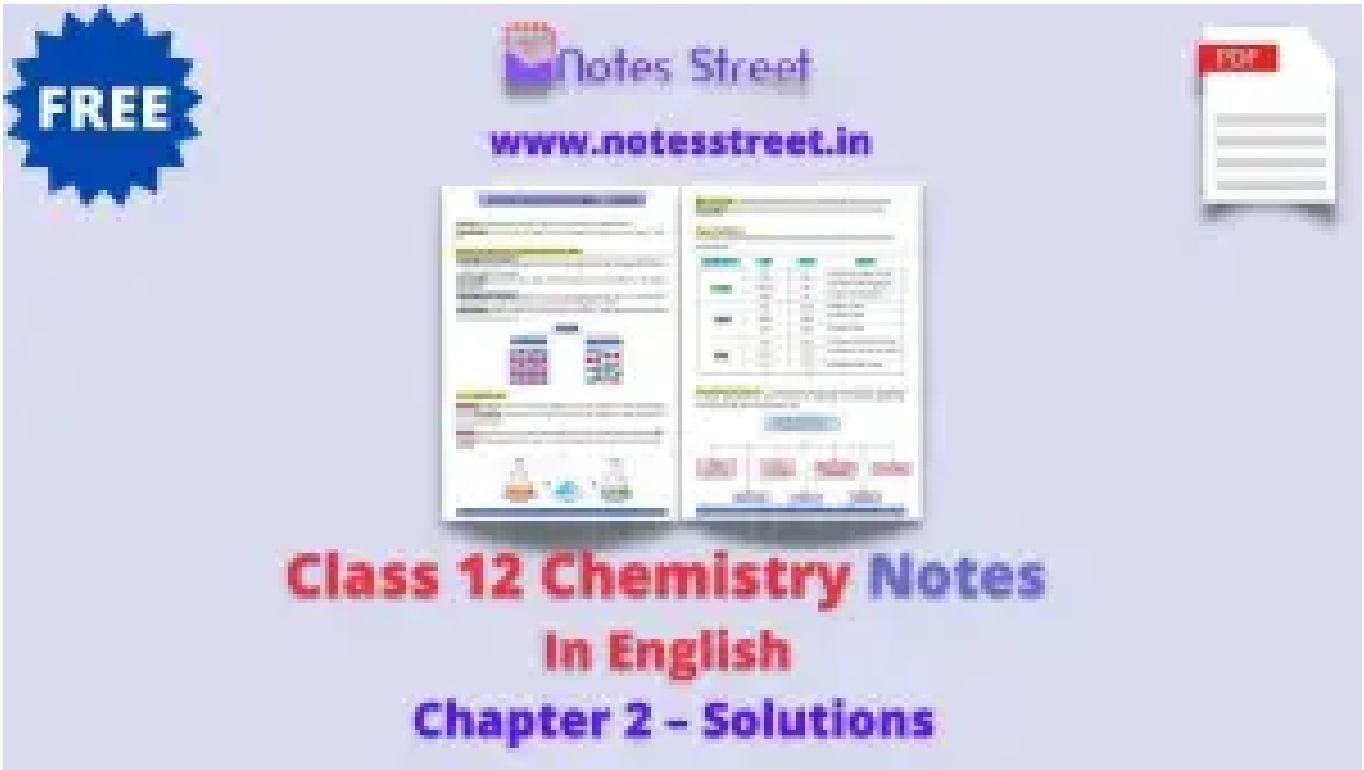


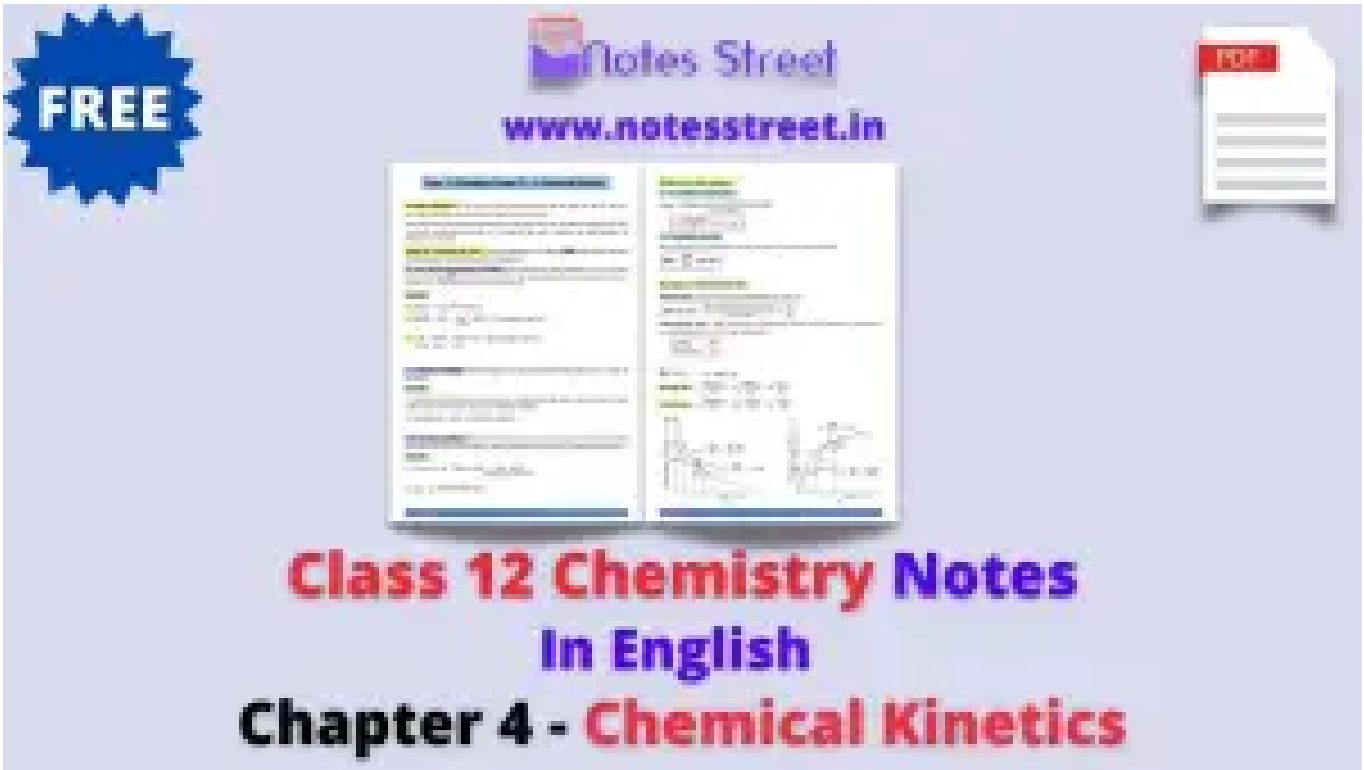
Class 12 Chemistry Notes Chapter 1 Solid State

June 21, 2020



Class 12 Chemistry notes Chapter 2 Solutions

June 1, 2022



Class 12 Chemistry Notes Chapter 4 Chemical Kinetics

January 31, 2023